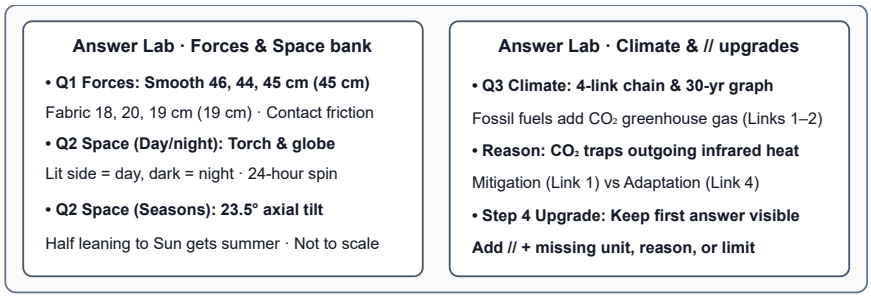
Reference sheets

Reference A · Answer Lab Bank Across Forces, Space, and Climate

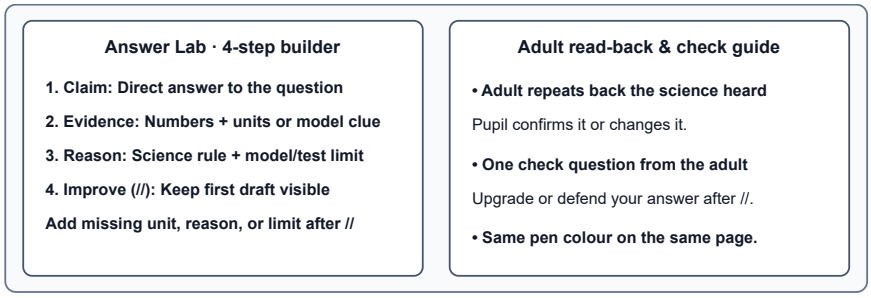


Answer Lab bank. It shows Claims, Evidence, and Reasons for Forces, Space, and Climate.

Reference only. Do not write on this sheet. Pick a question and use its Claim, Evidence, and Reason clues.

Question.	Step 1: Claim (Say it).	Step 2: Evidence (Support it).	Step 3: Science Reason & Limit.
1. Forces: Which surface lets the toy roll further?	The toy rolled further on smooth card than on rough fabric.	Smooth runs: 46, 44, 45 cm (mean 45 cm). Fabric runs: 18, 20, 19 cm (mean 19 cm).	Rough fabric has more contact friction opposing motion. Limit: 'in this test' (two surfaces).
2. Space: Why do we get day, night, and seasons?	Earth gets day and night every 24 hours, and tilted halves get seasons.	Torch-and-globe model: lit side faces torch; 23.5° tilted half leans to light.	24-hour spin gives day/night. 23.5° axial tilt during orbit gives seasons. Limit: 'not to scale'.
3. Climate: How does burning fossil fuels warm Earth?	Burning fossil fuels warms Earth's climate over 30+ years.	4-link warming chain and 30-year +1.2 °C record (not one day's weather).	Extra CO_2 greenhouse gas traps outgoing infrared heat. Mitigation (Link 1) vs adaptation (Link 4).

Reference B · Four-step Answer Lab builder and // upgrade guide



Four-step Answer Lab builder. It shows Claim, Evidence, Reason, and // upgrades.

Answer Lab step.	What to put in your answer.	Example from Question 1 (Forces).
1. Claim (Say it).	Give a clear, direct answer to the question.	'The toy rolled further on smooth card.'
2. Evidence (Support it).	Give the repeated numbers + units or the model clue.	'Smooth runs were 46, 44, and 45 cm (mean 45 cm) vs 18, 20, and 19 cm (mean 19 cm) on fabric.'
3. Reason & limit.	State the science mechanism and one model or test limit.	'Rough fabric made more contact friction pushing against the toy, in this test.'
4. Improve with //.	Keep your first answer visible. Add your upgrade after //.	'// Added cm units, the 45 cm vs 19 cm means, and contact friction.'

Made by Matt

Tier pack · Aspire

Use Reference A and Reference B. Work in your book or on this sheet. Point, speak, sign, draw, type, write, or tell an adult. You may pass.

Arrival · 4 questions

1. Use Box 2 of the arrival drawing (46, 44, 45 cm vs 18, 20, 19 cm). How do repeated trials strengthen the Evidence step?

Help. Think about reliability and mean values.

2. How does the Reason step turn a simple description into a Key Stage 3 explanation?

Help. Connect the clue to the underlying scientific mechanism.

3. Why should a strong Reason step also state a model or investigation limitation?

Help. Explain what the model or investigation cannot prove.

4. How does keeping your first draft before // demonstrate your scientific progress?

Help. Compare the draft before // with the edit after //.

You do tasks

1. Use Reference A and Reference B. Construct two Key Stage 3 explanations (Claim, Evidence, Reason, Limit) across two topics. **Staff I ☐ S ☐ C ☐**

Help. Include contact friction, 23.5° axial tilt, or CO_2 greenhouse gas and infrared heat.

2. Construct a third Key Stage 3 explanation for Climate. Link fossil fuel combustion, carbon dioxide, outgoing infrared radiation, and surface albedo. Compare Link 1 mitigation with Link 4 adaptation. **Staff I ☐ S ☐ C ☐**

Help. Examine Row 3 on Reference A.

3. Answer an adult check question on what your model or dataset cannot prove. Record your upgrade or justified defence after //. **Staff I ☐ S ☐ C ☐**

Help. Explain why 'in this test' or 'not to scale' makes a scientific claim honest.

Optional participation-loop prompt. Share one part of your science answer with an adult. Hear one check question or feedback step. Reply below or with // in your book. (Staff: the pupil responds to feedback with an adult, and R is marked only after the pupil has responded.)

Exit ticket

1. Compare the Evidence and Reason steps across Forces, Space, and Climate.

2. Think about an adult check question followed by a // edit or defence. Why is that part of working scientifically?

3. Give one example of a model limitation in Space and one test limitation in Forces.

Pupil voice

What helped you today, and what would you change?

Point, speak, sign, draw, write or choose. You may pass.

Feedback and response · Aspire

Feedback codes

Staff I ☐ S ☐ C ☐ · Route card feedback boxes: VF ☐ · WS ☐ · I ☐ · E ☐ · NS + … ☐ · R ☐ · // ☐ · ? ☐

Code	Meaning
VF	Verbal feedback
WS	Worked with support
I	Independent
NS + …	Next step
E	Evidence captured
R	Pupil responded to feedback
//	Self-edit point
?	Read this back to me

My response

My response: Record one response to feedback in your usual mode. You may pass.

Adult checks the response and ticks it, or writes R (Pupil responded to feedback).

Made by Matt